Jungi Lee

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RESEARCH INTERESTS

Computer Architecture, Accelerators for Emerging Workloads, Efficient AI Systems, Hardware-Software Co-Design

EDUCATION

Seoul National University *Seoul, Republic of Korea*

M.S./Ph.D. in Electrical and Computer Engineering, advised by Prof. Jaewoong Sim *Mar. 2023 – Present*

B.S. in Electrical and Computer Engineering *Mar. 2017 – Feb. 2023*

RESEARCH EXPERIENCE

Machine Learning Intern *May 2026 – Present*
NVIDIA (Mentor: Asit Mishra) *Santa Clara, CA, USA*

- **Revisiting NVFP4 design for efficient LLM pretraining, RL, and inference while maintaining accuracy**
  - Analyzed the end-to-end overhead of existing NVFP4 recipes in LLM pretraining, RL, and inference. Based on this analysis, redesigned NVFP4 with different block sizes, rounding, and scaling methods to reduce the overhead while keeping accuracy comparable to the existing recipes.

Graduate Research Assistant *Mar. 2023 – Present*
Seoul National University (Advisor: Prof. Jaewoong Sim) *Seoul, Republic of Korea*

- **MX+ (MICRO '25): Extending the MX formats to recover LLM accuracy**
  - Observed that the exponent of a block amax (the largest-magnitude element) is implied by the shared scale and does not need to be stored; repurposed the exponent bits of a block amax as extra mantissa bits for higher precision.
  - Integrated MX+ into CUTLASS and Triton, and proposed integration into Tensor Cores to further reduce overhead.
- **InfiniGen (OSDI '24): A KV cache management framework for offloading-based LLM serving systems**
  - Designed an attention speculation algorithm that amplifies a few channels in the query and key matrices and uses them to speculate attention scores with almost no overhead.
  - InfiniGen keeps the KV cache in CPU memory and prefetches only the essential entries using the algorithm, reducing transfer overhead and providing scalability under long-context generation.
- **Tender (ISCA '24): Algorithm-hardware co-design for efficient LLM inference**
  - Co-designed a channel grouping algorithm and hardware architecture to quantize activations and weights of LLMs down to INT4 without mixed-precision or training, with higher accuracy than state-of-the-art solutions.
  - Implemented Tender in SystemVerilog, achieving significant area reduction over state-of-the-art accelerators.
- **GSCore (ASPLOS '24): Architectural support for 3D Gaussian Splatting**
  - Participated in designing an accelerator that efficiently executes the rendering pipeline of 3D Gaussian Splatting.
  - Implemented an end-to-end cycle-level simulator in C++ with DRAM timing for functional and timing simulation.

PUBLICATION

* indicates equal contribution.

Soohyun Cha, **Jungi Lee**, and Jaewoong Sim, AdaTree: Serving-Aware Adaptive Tree Construction for Speculative Decoding, in Proceedings of the 40th Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2026

Junseo Lee, Sangyun Jeon, **Jungi Lee**, Junyong Park, and Jaewoong Sim, GRTX: Efficient Ray Tracing for 3D Gaussian-Based Rendering, in Proceedings of the 32nd International Symposium on High Performance Computer Architecture (**HPCA**), 2026

Jungi Lee, Junyong Park, Soohyun Cha, Jaehoon Cho, and Jaewoong Sim, MX+: Pushing the Limits of Microscaling Formats for Efficient Large Language Model Serving, in Proceedings of the 58th ACM/IEEE International Symposium on Microarchitecture (**MICRO**), 2025

Wonbeom Lee*, **Jungi Lee***, Junghwan Seo, and Jaewoong Sim, InfiniGen: Efficient Generative Inference of Large Language Models with Dynamic KV Cache Management, in Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation (**OSDI**), 2024

Jungi Lee*, Wonbeom Lee*, and Jaewoong Sim, Tender: Accelerating Large Language Models via Tensor Decomposition and Runtime Requantization, in Proceedings of the 51st ACM/IEEE International Symposium on Computer Architecture (**ISCA**), 2024

Junseo Lee, Seokwon Lee, **Jungi Lee**, Junyong Park, and Jaewoong Sim, GSCore: Efficient Radiance Field Rendering via Architectural Support for 3D Gaussian Splatting, in Proceedings of the 2024 International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), 2024

Junseo Lee, Kwanseok Choi, **Jungi Lee**, Seokwon Lee, Joonho Whangbo, and Jaewoong Sim, NeuRex: A Case for Neural Rendering Acceleration, in Proceedings of the 50th ACM/IEEE International Symposium on Computer Architecture (**ISCA**), 2023

TEACHING EXPERIENCE

Graduate Teaching Assistant *Mar. 2023 – Dec. 2023*
Seoul National University *Seoul, Republic of Korea*

- ECE 315.A - Digital Systems Design and Experiments Fall 2023
- ECE 322 - Computer Organization Spring 2023

Led lab sessions, answered questions about the class material, and graded labs/exams (ECE 315.A, ECE 322). Guided a course project on building a CNN accelerator on an FPGA (ECE 315.A).

PROFESSIONAL SERVICE

Reviewer of IEEE Transactions on Very Large Scale Integration Systems (TVLSI) *2026*

SKILL

Programming Language: C/C++, Python, CUDA, Verilog, SystemVerilog, Unix/Linux

Tools: PyTorch, Intel Pin, TVM